Finance – Questions & Solutions

November 2008

QUESTION 7

7.1 R1 570 is invested at 12% p.a. compound interest. After how many years will the investment be worth R23 000? (4)

7.2 A farmer has just bought a new tractor for R800 000. He has decided to replace the tractor in 5 years' time, when its trade-in value will be R200 000. The replacement cost of the tractor is expected to increase by 8% per annum.

7.2.1 The farmer wants to replace his present tractor with a new one in 5 years' time. The farmer wants to pay cash for the new tractor, after trading in his present tractor for R200 000. How much will he need to pay? (3)

7.2.2
- One month after purchasing his present tractor, the farmer deposited x rands into an account that pays interest at a rate of 12% p.a., compounded monthly.
- He continued to deposit the same amount at the end of each month for a total of 60 months.
- At the end of 60 months he has exactly the amount that is needed to purchase a new tractor, after he trades in his present tractor.

Calculate the value of x. (6)

7.2.3 Suppose that 12 months after the purchase of the present tractor and every 12 months thereafter, he withdraws R5 000 from his account, to pay for maintenance of the tractor. If he makes 5 such withdrawals, what will the new monthly deposit be? (4)

[17]



QUESTION 7

Penalise ONCE in question 7 for early rounding off.

7.1	$A = P(1+i)^n$ $23000 = 1570(1.12)^n$ $(1.12)^n = 14{,}64968153..$ $n\log(1{,}12) = \log 14{,}64968153..$ $n = 23{,}69$ years (23,68701...) or $n = 24$ years or $n = 23$ years 8 months or $n = 23{,}7$ years **OR** $A = P(1+i)^n$ $23000 = 1570(1+\frac{12}{100})^n$ $(1.12)^n = 14{,}64968153..$ $n\log(1{,}12) = \log 14{,}64968153..$ $n = 23{,}69$ years (23,68701...) or $n = 24$ years or $n = 23$ years 8 months or $n = 23{,}7$ years Note: Accept 24 years : 4 / 4 Incorrect Formula: 0/4	✓ formula ✓ substitution ✓ apply log function ✓answer (4) ✓ formula ✓ substitution of $\frac{12}{100}$ ✓ apply log function ✓answer (4)
7.2.1	$A = P(1+i)^n$ $= 800000(1.08)^5$ $= R1175462{,}46$ $\therefore R1175462{,}46 - R200\,000$ $= R975462{,}46$ Some calculators give R 975 462,50	✓ substitution ✓ R 1 175 462,46 ✓ R 975 462,46 (3) Incorrect Formula: 0/3
7.2.2	$F = \frac{x[(1+i)^n - 1]}{i}$ $975462{,}46 = x\frac{[1{,}01]^{60} - 1}{0{,}01}$ $\frac{975462{,}46 \times 0.01}{[1{,}01]^{60} - 1} = x$ $x = R\,11944{,}00$	✓ F = R975462,46 or answer in 7.2.1 ✓ $n = 60$ ✓ $i = 1{,}01$ ✓ formula ✓ simplification ✓ answer (6)

	OR $975462,46 = x\dfrac{[1,01]^{60}-1}{0,01}$ $975462,46 = 81,66966986x$ $x = R\,11944,00$	✓ F = R975462,46 ✓ $n = 60$ ✓ $i = 1,01$ ✓ formula ✓ simplification ✓ answer (6) Note: Continued Accuracy applies.
7.2.3	$Service = [5000(1,01)^{48} + 5000(1,01)^{36} + 5000(1,01)^{24} + 5000(1,01)^{12} + 5000]$ $= 32197,77$ $975462,46 = x\dfrac{[1,01]^{60}-1}{0,01} - Service$ $975462,46 = 81,66966986x - 32197,77$ $x = R\,12338,24$ **OR** $Service = \dfrac{5000[1,01^{60}-1]}{1,01^{12}-1}$ $= 32197,77$ $975462,46 = x\dfrac{[1,01]^{60}-1}{0,01} - Service$ $975462,46 = 81,66966986x - 32197,77$ $x = R\,12338,24$ **OR** Present Value payment of R 5000 $= 5000\{(1,01)^{-12} + (1,01)^{-24} + (1,01)^{-36} + (1,01)^{-48} + (1,01)^{-60}\}$ $= 5000(1,01)^{-12}\left\{\dfrac{1-(1,01)^{-60}}{1-(1,01)^{-12}}\right\}$ = R 17 723,25 Present Value of the sinking fund $= 975462,46(1,01)^{-60}$ = R 536 942,94 Total Value of sinking fund = R 17 723,25 + R 536 942,94 = R 554 666,19 $\therefore\ 554666,19 = x\left\{\dfrac{1-(1,01)^{-60}}{0,01}\right\}$ x = R 12 338,24 **OR**	✓✓ 32 197,77 ✓ setting up of correct equation ✓ answer (4) ✓✓ 32 197,77 ✓ setting up of correct equation ✓ answer (4) ✓ 17723,25 ✓ 554666,19 ✓ setting up of correct equation ✓ answer (4)

$(1+i_{eff}) = (1+0{,}01)^{12}$

$i_{eff} = 0{,}12682503.....$

$P(1+i)^n$

$= 5000\dfrac{(1{,}12682503)^5 - 1}{0{,}12682503}$

$= 32197{,}77$

✓ substitution into formula
✓ 32 197,77

$975462{,}46 = x\dfrac{[1{,}01]^{60} - 1}{0{,}01} - 32197{,}77$

$975462{,}46 = 81{,}66966986x - 32197{,}77$

$x = R\,12338{,}24$

✓ setting up of correct equation
✓ answer R 12 338,24
(4)

OR

$5000 = \dfrac{x[(1{,}01)^{12} - 1)}{0{,}01}$

$x = \dfrac{5000 \times 0{,}01}{1{,}01^{12} - 1}$

$x = 394{,}24$

So monthly deposit must be increased by R 394,24
New monthly deposit
= R 11 944 + R 394,24
= R 12 338,24

✓ substitution into formula
✓ 394,24

✓ setting up of correct equation

✓ answer R 12 338,24
(4)

[17]



November 2009

QUESTION 9

9.1 A photocopier valued at R24 000 depreciates at a rate of 18% p.a. on the reducing-balance method. After how many years will its value be R15 000? (4)

9.2 A car that costs R130 000 is advertised in the following way: 'No deposit necessary and first payment due three months after date of purchase.' The interest rate quoted is 18% p.a. compounded monthly.

9.2.1 Calculate the amount owing two months after the purchase date, which is one month before the first monthly payment is due. (3)

9.2.2 Herschel bought this car on 1 March 2009 and made his first payment on 1 June 2009. Thereafter he made another 53 equal payments on the first day of each month.

(a) Calculate his monthly repayments. (3)

(b) Calculate the total of all Herschel's repayments. (1)

9.2.3 Hashim also bought a car for R130 000. He also took out a loan for R130 000, at an interest rate of 18% p.a. compounded monthly. He also made 54 equal payments. However, he started payments one month after the purchase of the car. Calculate the total of all Hashim's repayments. (4)

9.2.4 Calculate the difference between Herschel's and Hashim's total repayments. (1)

[16]



QUESTION 9

Penalise ONCE in question 9 for early rounding off.

9.1	$A = P(1-i)^n$ $15000 = 24000(1-0{,}18)^n$ $0{,}625 = (0{,}82)^n$ $n = \frac{\log 0{,}625}{\log 0{,}82}$ $= 2{,}37$ years	**NOTE:** If subs A and P incorrectly: Answer would be $n = -2{,}37$ years $\therefore n = 2{,}37$ years: 2 / 4 If subs A and P incorrectly: Answer would be $n = -2{,}37$ years : 1 / 4 Answer $n = 2{,}4$ years 4 / 4 Answer rounded to 3 years and all calculations shown and $n = 2{,}37$ shown: 4/4 Answer rounded to 3 years and $n = 2{,}37$ not shown: 3 / 4	✓ substitution ✓ simplification ✓application of logs ✓ answer (4) Incorrect formula: 0/4
9.2.1	$130\,000\left(1+\frac{0{,}18}{12}\right)^2$ $= 130000\,(1{,}015)^2$ = R 133 929,25	**NOTE:** – 1 per error for incorrect substitution to a max of 2 marks	✓✓ substitution ✓ answer (3) Incorrect formula: 0/3



9.2.2(a)

$$133\,929{,}25 = \frac{x[1-(1{,}015)^{-54}]}{0{,}015}$$

$$2008{,}93875 = x[1-(1{,}015)^{-54}]$$

$$x = \text{R } 3636{,}36$$

✓ $n = 54$

✓ substitution of 133 929,25

✓ answer

(3)

OR

$$133\,929{,}25\left(1+\frac{0{,}18}{12}\right)^{54} = \frac{x\left[\left(1+\frac{0{,}18}{12}\right)^{54}-1\right]}{\frac{0{,}18}{12}}$$

$$299255{,}2087 = 82{,}29517136...x$$

$$x = \text{R } 3636{,}36$$

✓ $n = 54$

✓ substitution of 133 929,25

✓ answer

(3)

OR

$$130000\left(1+\frac{0{,}18}{12}\right)^{56} = \frac{x\left[\left(1+\frac{0{,}18}{12}\right)^{54}-1\right]}{\frac{0{,}18}{12}}$$

$$299255{,}2087 = 82{,}29517136...x$$

$$x = \text{R } 3636{,}36$$

✓ $n = 54$

✓ $130000\left(1+\frac{0{,}18}{12}\right)^{56}$

✓ answer

(3)

9.2.2(b)

$$\text{Total} = 3636{,}36 \times 54$$

$$= \text{R}196\,363{,}66$$

NOTE:
Accept answer = R 196

✓answer

(1)

9.2.3

$$130\,000 = \frac{x[1-(1{,}015)^{-54}]}{0{,}015}$$

$$1950 = x[1-(1{,}015)^{-54}]$$

$$x = \text{R } 3529{,}68$$

✓ 130 000
✓ $i = 0{,}015$

✓ answer 3529,68

$$\text{Total payments} = \text{R } 3529{,}68 \times 54$$

$$= \text{R } 190\,602{,}72$$

✓ answer R 190 602,72

(4)

OR

9.2.3 contd	$130\,000\left(1+\frac{0,18}{12}\right)^{54}=\frac{x\left[\left(1+\frac{0,18}{12}\right)^{54}-1\right]}{\frac{0,18}{12}}$ $290475,5842 = 82,29517136...x$ $x = \text{R } 3529,68$ Total payments = R 3529,68 × 54 = R 190 602,72 **OR** $130\,000\left(1+\frac{0,18}{12}\right)^{55}=\frac{x\left(1+\frac{0,18}{12}\right)\left[\left(1+\frac{0,18}{12}\right)^{54}-1\right]}{\frac{0,18}{12}}$ $290475,5842 = 82,29517136...x$ $x = \text{R } 3529,68$ Total payments = R 3529,68 × 54 = R 190 602,72 **NOTE:** Disregard the cents values if they are incorrect.	✓ $130\,000\left(1+\frac{0,18}{12}\right)^{54}$ ✓ $i = 0,015$ ✓ answer 3529,68 ✓ answer R 190 602,72 (4) ✓ $130\,000\left(1+\frac{0,18}{12}\right)^{55}$ ✓ $i = 0,015$ ✓ answer 3529,68 ✓ answer R 190 602,72 (4)
9.2.4	R196 363,66 – R190 602,72 =R5 760,96	✓ answer (1) **[16]**

February 2010

QUESTION 9

9.1 Andrew wants to borrow money to buy a motorbike that costs R55 000,00 and plans to repay the full amount over a period of 4 years in monthly instalments. He is presented with TWO options:

Option 1: *The bank calculates what Andrew would owe if he borrows R55 000,00 for 4 years at a simple interest rate of 12,75% p.a., and then pays that amount back in equal monthly instalments over 4 years.*

Option 2: *He borrows R55 000,00 from the bank. He pays the bank back in equal instalments over 4 years, the first payment being made at the end of the first month. Compound interest at 20% p.a. is charged on the reducing balance.*

9.1.1 If Andrew chooses Option 1, what will his monthly instalment be? (4)

9.1.2 Which option is the better option for Andrew? Justify your answer with appropriate calculations. (4)

9.1.3 What interest rate should replace 12,75% p.a. in Option 1 so that there is no difference between the two options? (3)

9.2 Lindiwe receives a bursary of R80 000,00 for her studies at university. She invests the money at a rate of 13,75% p.a., compounded yearly. She decides to withdraw R25 000,00 at the end of each year for her studies, starting at the end of the first year.

Determine for how many full years will this investment finance her studies. (4)

9.3 Given: $A = P(1+ni)$ where P and i are positive constants

9.3.1 State whether the graph of A, as a function of n, is linear, quadratic, exponential or none of these. (1)

9.3.2 Draw a possible graph of A, as a function of n. (2)

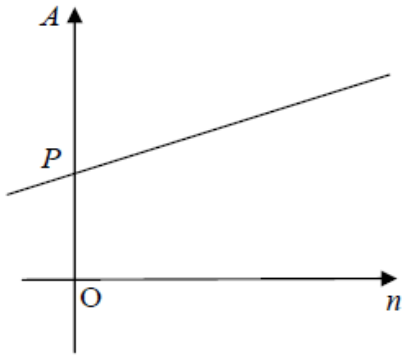
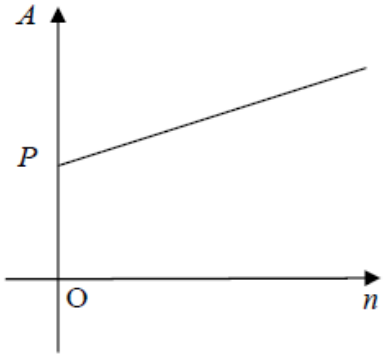
9.3.3 If n increases by 1, then determine the increase in A. (1)
[19]



QUESTION 9

9.1.1	Total amount $= P(1 + in)$ $= 55\,000(1 + 0{,}1275(4))$ $= 83\,050$ Monthly instalment $= \frac{83050}{4 \times 12}$ $= \text{R } 1730{,}21$	✓ substitution into simple interest formula ✓ answer ✓ ÷ 48 ✓ answer (4)
9.1.2	$55000 = \frac{x\left[1-\left(1+\frac{0{,}2}{12}\right)^{-12\times 4}\right]}{\frac{0{,}2}{12}}$ $x = \text{R } 1\,673{,}67$ a better option because monthly repayments are less.	✓ substitution into formula ✓ $i = \frac{0{,}2}{12}$ ✓ $n = 48$ ✓ answer (4)
9.1.3	$1673{,}67 \times 48$ $= 80336{,}16$ $80336{,}16 = 55000(1 + 4i)$ $1{,}460657455\ldots = 1 + 4i$ $i = 0{,}11516436\ldots$ Rate = 11,52%	✓ 80336,16 ✓ $80336{,}16 = 55000(1 + 4i)$ ✓ answer (3)
9.2	$80000 = \frac{25000\left[1-(1+0{,}1375)^{-n}\right]}{0{,}1375}$ $\frac{11}{25} = 1-(1+0{,}1375)^{-n}$ $\frac{14}{25} = (1+0{,}1375)^{-n}$ $\log\frac{14}{25} = -n\log(1{,}1375)$ $n = 4{,}50054779$ The money will last for 4 full years **OR** Candidate guesses 4 years. Then balance available at the end of 4 years (after the 4th withdrawal) is $80000(1+0{,}1375)^4 - 25000\left(\frac{(1+0{,}1375)^4-1}{0{,}1375}\right) = R11354{,}86.$ At the end of the 5th year cannot have grown to R25000.	✓ substitution into correct formula ✓ simplification ✓ taking log of both sides ✓ answer (4) ✓ guesses 4 years ✓✓ calculates balance at end of 4th year ✓ conclusion about balance. (4)



9.3.1	$A = P + (Pi)n$ which is a linear function of n.	✓ linear (1)
9.3.2	 Accept also: 	✓ $P > 0$ ✓ slope >0 (2)
9.3.3	The slope is Pi . Therefore this is the increase for A for each increase of 1 in n. **OR** $A(n+1) - A(n) = [P + Pi(n+1)] - [P + Pi\,n]$ $= Pi[(n+1) - n]$ $= Pi$ **OR** $A(1) - A(0) = (P + Pi) - (P + 0) = Pi$	✓ Pi (1) **[19]**

November 2010

QUESTION 7

7.1 At what annual percentage interest rate, compounded quarterly, should a lump sum be invested in order for it to double in 6 years? (5)

7.2 Timothy buys furniture to the value of R10 000. He borrows the money on 1 February 2010 from a financial institution that charges interest at a rate of 9,5% p.a. compounded monthly. Timothy agrees to pay monthly instalments of R450. The agreement of the loan allows Timothy to start paying these equal monthly instalments from 1 August 2010.

7.2.1 Calculate the total amount owing to the financial institution on 1 July 2010. (2)

7.2.2 How many months will it take Timothy to pay back the loan? (4)

7.2.3 What is the balance of the loan immediately after Timothy has made the 25th payment? (3)

[14]



QUESTION 7

7.1	$A = P(1+i)^n$ $2P = P\left(1+\frac{r}{4}\right)^{6\times4}$ $2 = \left(1+\frac{r}{4}\right)^{24}$ $1+\frac{r}{4} = 2^{\frac{1}{24}}$ $r = 4\left(2^{\frac{1}{24}}-1\right)$ $r = 4\left(2^{\frac{1}{24}}\right)-4$ $r = 0{,}1172\ldots$ rate = 11,72% p.a. compounded quarterly **OR** $A = P(1+i)^n$ $2P = P\left(1+\frac{r}{400}\right)^{6\times4}$ $2 = \left(1+\frac{r}{400}\right)^{24}$ $1+\frac{r}{400} = 2^{\frac{1}{24}}$ $r = 400\left(2^{\frac{1}{24}}-1\right)$ $r = 400\left(2^{\frac{1}{24}}\right)-400$ $r = 11{,}72\%$ p.a. **Note:** Penalty 1 for incorrect rounding off.	✓ 2P ✓ $\frac{r}{4}$ and 24 ✓ $1+\frac{r}{4} = 2^{\frac{1}{24}}$ ✓ $r = 4\left(2^{\frac{1}{24}}\right)-4$ ✓ answer (5) ✓ 2P ✓ $\frac{r}{400}$ and 24 ✓ $1+\frac{r}{400} = 2^{\frac{1}{24}}$ ✓ $r = 400\left(2^{\frac{1}{24}}\right)-400$ ✓ answer (5)
7.2.1	$A = 10000\left(1+\frac{0{,}095}{12}\right)^5$ = R 10 402,15	✓ substitution in correct formula ✓ answer (2)



7.2.2

$$10402{,}15 = \frac{450\left[1-\left(1+\frac{0{,}095}{12}\right)^{-n}\right]}{\frac{0{,}095}{12}}$$

✓ 10 402,15
✓ substitution into present value formula

$$0{,}183000787 = 1-\left(1+\frac{0{,}095}{12}\right)^{-n}$$

Note:
Incorrect Formula
No marks

$$\left(1+\frac{0{,}095}{12}\right)^{-n} = 0{,}816999213$$

$$\log\left(1+\frac{0{,}095}{12}\right)^{-n} = \log 0{,}816999213$$

✓ application of logs

$$-n\log\left(1+\frac{0{,}095}{12}\right) = \log 0{,}816999213\ldots$$

$$n = 25{,}63151282....$$

$$n = 25{,}63 \text{ months}$$

$$n = 26$$

✓ answer

Accept: $n = 31$ (because of first 5 months)

(4)

OR

$$10402{,}15\left(1+\frac{0{,}095}{12}\right)^{n} = \frac{450\left[\left(1+\frac{0{,}095}{12}\right)^{n}-1\right]}{\frac{0{,}095}{12}}$$

✓ 10 402,15
✓ substitution into future value formula

$$10402{,}15\left(1+\frac{0{,}095}{12}\right)^{n} = 56842{,}10526\left[\left(1+\frac{0{,}095}{12}\right)^{n}-1\right]$$

$$56842{,}10526 = 46439{,}95526\left(1+\frac{0{,}095}{12}\right)^{n}$$

$$\log 1{,}223991387 = n\log\left(1+\frac{0{,}095}{12}\right)$$

✓ application of logs

$$n = \frac{\log 1{,}223991387}{\log\left(1+\frac{0{,}095}{12}\right)}$$

$$n = 25{,}63 \quad \text{months}$$

$$n = 26$$

✓ answer

Accept: $n = 31$ (because of first 5 months)

(4)

Note: If the Present value of R 10 000 is used, then $n = 25{,}53$ months is obtained. Max 3 / 4 marks.

7.2.3

Balance outstanding after 25 months

$$= 10402{,}15\left(1+\frac{0{,}095}{12}\right)^{25} - \frac{450\left[\left(1+\frac{0{,}095}{12}\right)^{25}-1\right]}{\frac{0{,}095}{12}}$$

$= \text{R } 282{,}36$

✓ correct formula
✓ substitution into $\frac{450\left[\left(1+\frac{0{,}095}{12}\right)^{25}-1\right]}{\frac{0{,}095}{12}}$
✓ answer

(3)

OR

Balance Outstanding after 25 months

$$= 10000\left(1+\frac{0{,}095}{12}\right)^{30} - \frac{450\left[\left(1+\frac{0{,}095}{12}\right)^{25}-1\right]}{\frac{0{,}095}{12}}$$

$= \text{R } 282{,}36$

✓ correct formula
✓ $\frac{450\left[\left(1+\frac{0{,}095}{12}\right)^{25}-1\right]}{\frac{0{,}095}{12}}$
✓ answer

(3)

OR

$n = 25{,}6315128204.... - 25$

$= 0{,}6315128204 ...$

Balance Outstanding after 25 months

$$= \frac{450\left[1-\left(1+\frac{0{,}095}{12}\right)^{-0{,}631512804}\right]}{\frac{0{,}095}{12}}$$

$= \text{R } 282{,}36$

Note: Accept
If a candidate uses – 0,63, the final answer is R 281,68

✓ correct formula
✓ substitution into $\frac{450\left[1-\left(1+\frac{0{,}095}{12}\right)^{-0{,}631512804}\right]}{\frac{0{,}095}{12}}$
✓ answer

(3)

OR

Present value at beginning of 25 months

$$= 10402{,}15 - \frac{450\left[1-\left(1+\frac{0{,}095}{12}\right)^{-25}\right]}{\frac{0{,}095}{12}}$$

$= \text{R } 231{,}84$

Balance Outstanding

$$= 231{,}84\left(1+\frac{0{,}095}{12}\right)^{25}$$

$= \text{R } 282{,}36$

✓ correct formula
✓ substitution into $\frac{450\left[1-\left(1+\frac{0{,}095}{12}\right)^{-25}\right]}{\frac{0{,}095}{12}}$
✓ answer

(3)

[14]



February 2011

QUESTION 8

8.1 R1 430,77 was invested in a fund paying i% p.a. compounded monthly. After 18 months the fund had a value of R1 711,41. Calculate i. (4)

8.2 A father decided to buy a house for his family for R800 000. He agreed to pay monthly instalments of R10 000 on a loan which incurred interest at a rate of 14% p.a. compounded monthly. The first payment was made at the end of the first month.

8.2.1 Show that the loan would be paid off in 234 months. (4)

8.2.2 Suppose the father encountered unexpected expenses and was unable to pay any instalments at the end of the 120th, 121st, 122nd and 123rd months. At the end of the 124th month he increased his payment so as to still pay off the loan in 234 months by 111 equal monthly payments.
Calculate the value of this new instalment. (7)
[15]

QUESTION 8

8.1	$A = P(1+i)^n$ $1711,41 = 1430,77\left(1+\frac{i}{12}\right)^{18}$ $\left(1+\frac{i}{12}\right)^{18} = 1,196146131...$ **OR** $\left[\frac{1711,41}{1430,77}\right]^{\frac{1}{18}} = 1,00999...$ $1+\frac{i}{12} = 1,009999937...$ $\therefore i = 12(1,01-1)$ $i = 0,1199992431...$ $= 0,12$ Rate = 12, 00% p.a. compounded monthly. $= 12\%$	✓ substitution ✓ $\left(1+\frac{i}{12}\right)^{18} = 1,196146131...$ ✓ $1+\frac{i}{12} = 1,009999937...$ ✓ answer (4)

8.2.1

$$P_v = \frac{x\left[1-(1+i)^{-n}\right]}{i}$$

$$800000 = \frac{10000\left[1-\left(1+\frac{0,14}{12}\right)^{-n}\right]}{\frac{0,14}{12}}$$

✓ substitute into P_v

✓ $i = \frac{0,14}{12}$

$$1-\left(1+\frac{0,14}{12}\right)^{-n} = \frac{14}{15} \quad (= 0,933333)$$

$$\left(1+\frac{0,14}{12}\right)^{-n} = \frac{1}{15} \quad (= 0,06666666)$$

$$\log\left(1+\frac{0,14}{12}\right)^{-n} = \log\frac{1}{15}$$

✓ using logs

$$-n\log\left(1+\frac{0,14}{12}\right) = \log\frac{1}{15} \qquad \left(-n = \frac{\log\frac{1}{15}}{\log\left(1+\frac{0,14}{12}\right)} = -233,47\right)$$

$n = 233,47$

✓ answer

$\therefore$ the loan will be paid off at the end of the 234th month (4)

OR

Balance outstanding after 233rd month

$$= 800000\left(1+\frac{0,14}{12}\right)^{233} - \frac{10000\left[\left(1+\frac{0,14}{12}\right)^{233}-1\right]}{\frac{0,14}{12}}$$

✓ substitution into P formula

✓ 234

= R4 660,04 which is less than R10 000

Therefore the loan will be paid off after 234 months.

✓ answer

✓ argument

(4)

OR

Total value of the loan after 234 payments

$$= \frac{10000\left(1-\left(1+\frac{0,14}{12}\right)^{-234}\right)}{\frac{0,14}{12}}$$

✓ substitution into F formula

✓ 234

= R800 350,21

> R800 000 and the differences is less than R10 000

Therefore the loan will be paid off after 234 months.

✓ answer

✓ argument

(4)

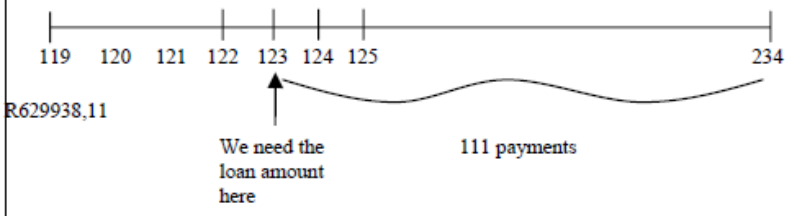


8.2.2

Balance Outstanding after 119 months

$$= 800000\left(1+\frac{0,14}{12}\right)^{119} - \frac{10000\left[\left(1+\frac{0,14}{12}\right)^{119}-1\right]}{\frac{0,14}{12}}$$

$$= \text{R}629\,938,11$$



Total Payable at the end of the 123rd month

$$= 629\,938,11\left(1+\frac{0,14}{12}\right)^{4}$$

$$= \text{R}659\,853,68$$

New instalment:

$$659\,853,68 = \frac{x\left[1-\left(1+\frac{0,14}{12}\right)^{-111}\right]}{\frac{0,14}{12}}$$

$$x = \text{R}10\,632,39$$

✓ $800000\left(1+\frac{0,14}{12}\right)^{119}$

✓ $\frac{10000\left[\left(1+\frac{0,14}{12}\right)^{119}-1\right]}{\frac{0,14}{12}}$

✓ R629 938,11

✓ $629938,11\left(1+\frac{0,14}{12}\right)^{4}$

✓ R659 853,68

✓ substitution into P_v

✓ answer

(7)

[15]

November 2011

QUESTION 7

7.1 How many years will it take for an article to depreciate to half its value according to the reducing-balance method at 7% per annum? (4)

7.2 Two friends each receive an amount of R6 000 to invest for a period of 5 years. They invest the money as follows:

- Radesh: 8,5% per annum simple interest. At the end of the 5 years, Radesh will receive a bonus of exactly 5% of the principal amount.
- Thandi: 8% per annum compounded quarterly.

Who will have the bigger investment after 5 years? Justify your answer with appropriate calculations. (6)

7.3 Nicky opened a savings account with a single deposit of R1 000 on 1 April 2011. She then makes 18 monthly deposits of R700 at the end of every month. Her first payment is made on 30 April 2011 and her last payment on 30 September 2012. The account earns interest at 15% per annum compounded monthly.

Determine the amount that should be in her savings account immediately after her last deposit is made (that is on 30 September 2012). (6)

[16]

QUESTION 7

7.1

$A = P(1-i)^n$

$\frac{P}{2} = P(1-0,07)^n$

$\frac{1}{2} = 0,93^n$

$\log\frac{1}{2} = n\log 0,93$

$n = \frac{\log\frac{1}{2}}{\log 0,93}$

$= 9,55$ years

OR

$A = P(1-i)^n$

$\frac{P}{2} = P(1-0,07)^n$

$\frac{1}{2} = 0,93^n$

$\log_{0,93}\frac{1}{2} = n$

$n = 9,55$ years

✓ $A = \frac{P}{2}$

✓ subs into correct formula

✓ log

✓ answer (4)

Note:
If candidate interchanges A and P
i.e. uses $P = \frac{A}{2}$: max 2/4 marks

Note:
If candidate uses incorrect formula: max 1/4 marks
for $A = \frac{P}{2}$

7.2	**Radesh:** $A = P(1+in)$ $= 6\,000(1+0{,}085\times 5)$ OR $= 8\,550$ $A = 6\,000 + 8{,}5\%\text{ of }6000\times 5$ $= 6000 + 510\times 5$ $= 6000 + 2550$ $= 8\,550$ Bonus $= 0{,}05\times 6\,000$ $= 300$ Received $= 8\,550 + 300$ $= \text{R}8\,850$ **Thandi:** $A = P(1+i)^n$ $= 6\,000\left(1+\frac{0{,}08}{4}\right)^{20}$ $= \text{R}8\,915{,}68$ Thandi's investment is bigger.	✓ 8 550 ✓ **R**8 850 ✓ $n = 20$ ✓ $i = \frac{0{,}08}{4}$ ✓ answer ✓ choice made (6)
7.3	F_v = initial deposit with interest + annuity $= 1\,000\left(1+\frac{0{,}15}{12}\right)^{18} + 700\left(\frac{\left(1+\frac{0{,}15}{12}\right)^{18}-1}{\frac{0{,}15}{12}}\right)$ $= 1\,250{,}58 + 14\,032{,}33$ $= \text{R}15\,282{,}91$ **OR** F_v = initial deposit with interest + annuity $= 1\,000\left(1+\frac{0{,}15}{12}\right)^{18} + 700\left(\frac{1-\left(1+\frac{0{,}15}{12}\right)^{-18}}{\frac{0{,}15}{12}}\right)\left(1+\frac{0.15}{12}\right)^{18}$ $= 1\,250{,}58 + 11220{,}68\left(1+\frac{0.15}{12}\right)^{18}$ $= 1\,250{,}58 + 14\,032{,}33$ $= \text{R}15\,282{,}91$	✓ $i = \frac{0{,}15}{12}$ or $\frac{1}{80}$ or 0,0125 ✓ $n = 18$ ✓ $n = 18$ ✓ $1\,000\left(1+\frac{0{,}15}{12}\right)^{18}$ ✓ $700\left(\frac{\left(1+\frac{0{,}15}{12}\right)^{18}-1}{\frac{0{,}15}{12}}\right)$ ✓ answer (6) ✓ $i = \frac{0{,}15}{12}$ or $\frac{1}{80}$ or 0,0125 ✓ $n = 18$ ✓ $n = 18$ ✓ $1\,000\left(1+\frac{0{,}15}{12}\right)^{18}$ ✓ $700\left(\frac{1-\left(1+\frac{0{,}15}{12}\right)^{-18}}{\frac{0{,}15}{12}}\right)\left(1+\frac{0.15}{12}\right)^{18}$ ✓ answer (6)

	OR $F_v = 300\left(1+\frac{0,15}{12}\right)^{18} + 700\left(\frac{\left(1+\frac{0,15}{12}\right)^{19}-1}{\frac{0,15}{12}}\right)$ $= 375,17 + 14\,907,74$ $= \text{R}15\,282,91$	✓ $i = \frac{0,15}{12}$ or $\frac{1}{80}$ or 0,0125 ✓ $n = 19$ (corresponding to 700) ✓ $n = 18$ (corresponding to 300) ✓ $300\left(1+\frac{0,15}{12}\right)^{18}$ ✓ $700\left(\frac{\left(1+\frac{0,15}{12}\right)^{19}-1}{\frac{0,15}{12}}\right)$ ✓ answer (6) **[16]**



February 2012

QUESTION 7

7.1 Lerato wants to purchase a house that costs R850 000. She is required to pay a 12% deposit and she will borrow the balance from a bank. Calculate the amount that Lerato must borrow from the bank. (2)

7.2 The bank charges interest at 9% per annum, compounded monthly on the loan amount. Lerato works out that the loan will carry an effective interest rate of 9,6% per annum. Is her calculation correct or not? Justify your answer with appropriate calculations. (4)

7.3 Lerato takes out a loan from the bank for the balance of the purchase price and agrees to pay it back over 20 years. Her repayments start one month after her loan is granted. Determine her monthly instalment if interest is charged at 9% per annum compounded monthly. (4)

7.4 Lerato can afford to repay R7 000 per month. How long will it take her to repay the loan amount if she chooses to pay R7 000 as a repayment every month? (4)

[14]



QUESTION 7

7.1	$\frac{88}{100} \times 850\,000$ $= R748\,000$ **OR** $\frac{12}{100} \times 850\,000$ $= R102\,000$ Loan amount $= R850\,000 - R102\,000$ $= R748\,000$	✓ $\frac{88}{100} \times 850\,000$ ✓ answer (2) ✓ $\frac{12}{100} \times 850\,000$ ✓ answer (2)
7.2	$1 + i_e = \left(1 + \frac{0,09}{12}\right)^{12}$ $i_e = 0,09380689$ $r = 9,38\%\ p.a$ $\neq 9,6\%$ Not correct **OR** $1 + 0,096 = \left(1 + \frac{i}{12}\right)^{12}$ $\sqrt[12]{1,096} = 1 + \frac{i}{12}$ $1,007668183 = 1 + \frac{i}{12}$ $i = 0,092018201$ $r = 9,2\%\ p.a$ $\neq 9\%$ Not correct	✓ $\frac{0,09}{12}$ ✓ substitution ✓ answer ✓ decision (4) ✓ $\frac{i}{12}$ ✓ substitution ✓ answer ✓ decision (4)
7.3	$P_v = \frac{x[1-(1+i)^{-n}]}{i}$ $748\,000 = \frac{x\left[1-\left(1+\frac{0,09}{12}\right)^{-240}\right]}{\frac{0,09}{12}}$ $x = R6\ 729,95$	✓ subs into correct formula ✓ $i = \frac{0,09}{12}$ ✓ n = – 240 ✓ answer (4)



7.4

$$P_v = \frac{x[1-(1+i)^{-n}]}{i}$$

$$748\,000 = \frac{7\,000\left[1-\left(1+\frac{0,09}{12}\right)^{-n}\right]}{\frac{0,09}{12}}$$

✓ subs into correct formula

$$\frac{561}{700} = 1-\left(1+\frac{0,09}{12}\right)^{-n}$$

$$\left(1+\frac{0,09}{12}\right)^{-n} = \frac{139}{700}$$

✓ simplification

$$-n\log\left(1+\frac{0,09}{12}\right) = \log\frac{139}{700}$$

✓ use of logs

$$n = 216,35 \text{ months}$$

$$= 18,03 \text{ years}$$

✓ answer

(4)

OR

$$P_v = \frac{x[1-(1+i)^{-12n}]}{i}$$

$$748\,000 = \frac{7\,000\left[1-\left(1+\frac{0,09}{12}\right)^{-12n}\right]}{\frac{0,09}{12}}$$

✓ subs into correct formula

$$\frac{561}{700} = 1-\left(1+\frac{0,09}{12}\right)^{-12n}$$

$$\left(1+\frac{0,09}{12}\right)^{-12n} = \frac{139}{700}$$

✓ simplification

$$-12n\log\left(1+\frac{0,09}{12}\right) = \log\frac{139}{700}$$

✓ use of logs

$$n = 18,03 \text{ years}$$

✓ answer

(4)

[14]

November 2012

QUESTION 7

7.1 A business buys a machine that costs R120 000. The value of the machine depreciates at 9% per annum according to the diminishing-balance method.

7.1.1 Determine the scrap value of the machine at the end of 5 years. (3)

7.1.2 After five years the machine needs to be replaced. During this time, inflation remained constant at 7% per annum. Determine the cost of the new machine at the end of 5 years. (3)

7.1.3 The business estimates that it will need R90 000 by the end of five years. A sinking fund for R90 000, into which equal monthly instalments must be paid, is set up. Interest on this fund is 8,5% per annum, compounded monthly. The first payment will be made immediately and the last payment will be made at the end of the 5-year period.

Calculate the value of the monthly payment into the sinking fund. (5)

7.2 Lorraine receives an amount of R900 000 upon her retirement. She invests this amount immediately at an interest rate of 10,5% per annum, compounded monthly.

She needs an amount of R18 000 per month to maintain her current lifestyle. She plans to withdraw the first amount at the end of the first month.

For how many months will she be able to live from her investment? (6)

[17]



QUESTION 7

7.1.1	$A = P(1-i)^n$ $= 120\,000(1-0,09)^5$ $= R\,74\,883,86$ **NOTE:** Incorrect formula (in 7.1.1 or 7.1.2) award max 1/3 marks	✓ i, n and P identified ✓ subs into correct formula ✓ answer (3)
7.1.2	$A = P(1+i)^n$ $= 120\,000(1+0,07)^5$ $= R168\,306,21$	✓ i, n and P identified ✓ subs into correct formula ✓ answer (3)
7.1.3	Sinking fund needed: $F_v = R\,90\,000$ $F_v = \frac{x[(1+i)^n - 1]}{i}$ $90\,000 = \frac{x\left[\left(1+\frac{0,085}{12}\right)^{61} - 1\right]}{\frac{0,085}{12}}$ $x = R\,1\,184,68$ **NOTE:** Incorrect formula award max 2/5 marks	✓ $F_v = R\,90\,000$ ✓ $i = \frac{0,085}{12} = \frac{17}{2400}$ in annuity formula ✓ $n = 61$ ✓ subs into correct formula ✓ answer (5)



OR

Consider the scenario as money deposited at the beginning of every month, but in the last month an additional payment was made at the end of the month:

$$F_v = \frac{x(1+i)\left[(1+i)^n - 1\right]}{i} + x$$

$$= x\left(\frac{(1+i)\left[(1+i)^n - 1\right]}{i} + 1\right)$$

$$90\,000 = x\left(\frac{\left(1+\frac{0{,}085}{12}\right)\left[\left(1+\frac{0{,}085}{12}\right)^{60} - 1\right]}{\frac{0{,}085}{12}} + 1\right)$$

$$x = \frac{90\,000\left(\frac{0{,}085}{12}\right)}{\left(1+\frac{0{,}085}{12}\right)\left[\left(1+\frac{0{,}085}{12}\right)^{60} - 1\right] + \left(\frac{0{,}085}{12}\right)}$$

$$= \text{R}1184,68$$

✓ $i = \frac{0{,}085}{12} = \frac{17}{2400}$ in annuity formula

✓ $n = 60$ in annuity formula

✓ $F_v = \text{R}\,90\,000$

✓ subs into correct formula

✓ answer

(5)

OR

Present value of sinking fund needed:

$$90\,000 = P_v\left(1+\frac{0{,}085}{12}\right)^{61}$$

$$P_v = \text{R}58\,513,03$$

Using the present value formula:

$$P_v = \frac{x[1-(1+i)^{-n}]}{i}$$

$$58\,513,03 = \frac{x\left[1-\left(1+\frac{0{,}085}{12}\right)^{-61}\right]}{\frac{0{,}085}{12}}$$

$$x = \text{R}\,1\,184,68$$

✓ $i = \frac{0{,}085}{12} = \frac{17}{2400}$ in annuity formula

✓ $n = 61$

✓ $P_v = \text{R}58\,513,03$

✓ subs into correct formula

✓ answer

(5)



7.2

$$P_v = \frac{x[1-(1+i)^{-n}]}{i}$$

$$900\,000 = \frac{18\,000\left[1-\left(1+\frac{0,105}{12}\right)^{-n}\right]}{\frac{0,105}{12}}$$

$$1-\frac{900\,000\left(\frac{0,105}{12}\right)}{18\,000} = \left(1+\frac{0,105}{12}\right)^{-n}$$

$$-n = \log_{\left(1+\frac{0,105}{12}\right)}\frac{9}{16}$$

$$n = 66,04 \text{ months}$$

She will be able to maintain her current lifestyle for a little more than 66 months using her pension money.

Note: If F_v formula used, possibly award one each for x, i, use of logs: max 3/6 marks

If any other incorrect formula is used, award 0/6 marks

✓ $x = 18\,000$
✓ $i = \frac{0,105}{12}$ in annuity formula
✓ subs into correct formula
✓ simplification
✓ use of logs
✓ answer in months
(6)

OR

$$P_v = \frac{x[1-(1+i)^{-n}]}{i}$$

$$900\,000 = \frac{18\,000\left[1-\left(1+\frac{0,105}{12}\right)^{-n}\right]}{\frac{0,105}{12}}$$

$$1-\frac{900\,000\left(\frac{0,105}{12}\right)}{18\,000} = \left(1+\frac{0,105}{12}\right)^{-n}$$

$$-n\log\left(1+\frac{0,105}{12}\right) = \log\frac{9}{16}$$

$$n = 66,04 \text{ months}$$

She will be able to maintain her current lifestyle for a little more than 66 months using her pension money.

Note: If candidate rounds off early in Question 7.2 (and obtain 58 months), penalise 1 mark

✓ $x = 18\,000$
✓ $i = \frac{0,105}{12}$ in annuity formula
✓ subs into correct formula
✓ simplification
✓ use of logs
✓ answer in months
(6)

OR

$$A = F_v$$

$$P(1+i)^n = \frac{x\left[(1+i)^n - 1\right]}{i}$$

$$900\,000\left(1+\frac{0{,}105}{12}\right)^n = \frac{18000\left[\left(1+\frac{0{,}105}{12}\right)^n - 1\right]}{\frac{0{,}105}{12}}$$

$$\frac{0{,}105}{12}\times 900\,000\left(1+\frac{0{,}105}{12}\right)^n = 18000\left(1+\frac{0{,}105}{12}\right)^n - 18000$$

$$18000 = 18000\left(1+\frac{0{,}105}{12}\right)^n - \frac{0{,}105}{12}\times 900\,000\left(1+\frac{0{,}105}{12}\right)^n$$

$$18000 = \left(1+\frac{0{,}105}{12}\right)^n\left(18000 - \frac{0{,}105}{12}\times 900\,000\right)$$

$$18000 \div \left(18000 - \frac{0{,}105}{12}\times 900\,000\right) = \left(1+\frac{0{,}105}{12}\right)^n$$

$$\frac{16}{9} = \left(1+\frac{0{,}105}{12}\right)^n$$

$$-n = \log_{\left(1+\frac{0{,}105}{12}\right)}\frac{9}{16}$$

$$n = 66{,}04 \text{ months}$$

She will be able to maintain her current lifestyle for a little more than 66 months using her pension money.

✓ $x = 18\,000$
✓ $i = \frac{0{,}105}{12}$ in annuity formula
✓ subs into correct formula

✓ simplification

✓ use of logs

✓ answer in months

(6)
[17]

February 2013

QUESTION 7

7.1 Raeesa invests R4 million into an account earning interest of 6% per annum, compounded annually. How much will her investment be worth at the end of 3 years? (3)

7.2 Joanne invests R4 million into an account earning interest of 6% per annum, compounded monthly.

7.2.1 She withdraws an allowance of R30 000 per month. The first withdrawal is exactly one month after she has deposited the R4 million. How many such withdrawals will Joanne be able to make? (6)

7.2.2 If Joanne withdraws R20 000 per month, how many withdrawals will she be able to make? (3)

[12]

QUESTION 8

Jeffrey invests R700 per month into an account earning interest at a rate of 8% per annum, compounded monthly. His friend also invests R700 per month and earns interest compounded semi-annually (that is every six months) at r% per annum. Jeffrey and his friend's investments are worth the same at the end of 12 months. Calculate r. **[3]**



QUESTION 7

7.1	$F = P(1+i)^n$ $= 4\,000\,000(1+0{,}06)^3$ $= \text{R}4\,764\,064$	✓ formula ✓ substitution ✓ answer (3)
7.2.1	$4\,000\,000 = \dfrac{30\,000\left[1-\left(1+\dfrac{0{,}06}{12}\right)^{-n}\right]}{\dfrac{0{,}06}{12}}$ $\dfrac{4000000\times\left(\dfrac{0{,}06}{12}\right)}{30000} = 1-\left(1+\dfrac{0{,}06}{12}\right)^{-n}$ $\dfrac{1}{3} = \left(1+\dfrac{0{,}06}{12}\right)^{-n}$ $\log_{\left(1+\frac{0{,}06}{12}\right)}\dfrac{1}{3} = -n$ $n = 220{,}27$ Therefore she will make 220 withdrawals of R30 000. **OR** $4\,000\,000 = \dfrac{30\,000\left[1-\left(1+\dfrac{0{,}06}{12}\right)^{-n}\right]}{\dfrac{0{,}06}{12}}$ $\dfrac{4000000\times\left(\dfrac{0{,}06}{12}\right)}{30000} = 1-\left(1+\dfrac{0{,}06}{12}\right)^{-n}$ $\dfrac{1}{3} = \left(1+\dfrac{0{,}06}{12}\right)^{-n}$ $\log\dfrac{1}{3} = -n\log\left(1+\dfrac{0{,}06}{12}\right)$ $n = 220{,}27$ Therefore she will make 220 withdrawals of R30 000.	✓ formula ✓ $i = \dfrac{0{,}06}{12}$ ✓ substitution into correct formula ✓ $\dfrac{1}{3} = \left(1+\dfrac{0{,}06}{12}\right)^{-n}$ ✓ correct use of logs ✓ answer of 220 withdrawals (6) ✓ formula ✓ $i = \dfrac{0{,}06}{12}$ ✓ substitution into correct formula ✓ $\dfrac{1}{3} = \left(1+\dfrac{0{,}06}{12}\right)^{-n}$ ✓ correct use of logs ✓ answer of 220 withdrawals (6)

7.2.2	$4\,000\,000 = \dfrac{20\,000\left[1-\left(1+\dfrac{0,06}{12}\right)^{-n}\right]}{\dfrac{0,06}{12}}$ $0 = \left(1+\dfrac{0,06}{12}\right)^{-n}$ She can make as many withdrawals as she pleases.	✓ $4\,000\,000 = \dfrac{20\,000\left[1-\left(1+\dfrac{0,06}{12}\right)^{-n}\right]}{\dfrac{0,06}{12}}$ ✓ $0 = \left(1+\dfrac{0,06}{12}\right)^{-n}$ ✓ conclusion (3) **[12]**

QUESTION 8

	$\left(1+\dfrac{0,08}{12}\right)^{12} = \left(1+\dfrac{r}{2}\right)^{2}$ $\dfrac{r}{2} = 0,040672622$ $r = 8,13452446\%$ $r = 8,13\%$	✓$\left(1+\dfrac{0,08}{12}\right)^{12}$ ✓$\left(1+\dfrac{i}{2}\right)^{2}$ ✓ answer **[3]**



November 2013

QUESTION 7

7.1 Mpho invests R12 500 for exactly k years. She earns interest at a rate of 9% per annum, compounded quarterly. At the end of k years, her investment is worth R30 440.

7.1.1 Calculate the effective annual interest rate of Mpho's investment. (2)

7.1.2 Determine the value of k. (5)

7.2 Darrel is planning to buy his first home. The bank will allow him to use a maximum of 30% of his monthly salary to repay the bond.

7.2.1 Calculate the maximum amount that the bank will allow Darrel to spend each month on his bond repayments, if Darrel earns R18 480 per month. (1)

7.2.2 Suppose, at the end of each month, Darrel repays the maximum amount allowed by the bank. How much money does Darrel borrow if he takes 25 years to repay the loan at a rate of 8% p.a., compounded monthly? (The first repayment is made one month after the loan is granted.) (4)

[12]



QUESTION/*VRAAG* 7

7.1.1

$$1+i_{eff}=\left(1+\frac{0,09}{4}\right)^4$$

$$i_{eff}=\left(1+\frac{0,09}{4}\right)^4-1$$

$$=0,093083318$$

$$=0,0931$$

$$=9,31\%$$

✓ $\left(1+\frac{0,09}{4}\right)^4$

✓ 9,31% or 0,0931

(2)

7.1.2

$$A=P(1+i)^n$$

$$30\,440=12\,500\left(1+\frac{0,09}{4}\right)^{4k}$$

$$\frac{30440}{12500}=\left(1+\frac{0,09}{4}\right)^{4k}$$

$$2,4352=1,0225^{4k}$$

$$4k=\log_{1,0225}2,4352$$

$$4k=40,00020365...$$

$$k=10 \text{ years}$$

$$A=P(1+i)^n$$

$$30\,440=12\,500\left(1+\frac{0,09}{4}\right)^{4k}$$

$$\frac{30440}{12500}=\left(1+\frac{0,09}{4}\right)^{4k}$$

$$2,4352=1,0225^{4k}$$

$$4k\log 1,0225=\log 2,4352$$

$$4k=\frac{\log 2,4352}{1,0225}$$

$$4k=40,00020365...$$

$$k=10 \text{ years}$$

✓ $n=4k$

✓ $i=\frac{0,09}{4}$

✓ subs into correct formula

✓ use of logs

✓ answer

(5)

OR

$$A=P(1+i)^n$$

$$30\,440=12\,500(1+0,09308...)^k$$

$$\frac{30440}{12500}=(1+0,09308...)^k$$

$$2,4352=(1,09308...)^k$$

$$k=\log_{1,09308...}2,4352$$

$$k=9,998336572...$$

$$k=10 \text{ years}$$

$$A=P(1+i)^n$$

$$30\,440=12\,500(1+0,09308...)^k$$

$$\frac{30440}{12500}=(1+0,09308....)^k$$

$$2,4352=(1,09308....)^k$$

$$k\log 1,09308....=\log 2,4352$$

$$k=\frac{\log 2,4352}{\log 1,09308...}$$

$$k=9,998336572...$$

$$k=10 \text{ years}$$

✓ $n=k$

✓ i=0,09308… (from 7.1.1)

✓ subs into correct formula

✓ use of logs

OR

NOTE:
Incorrect formula: max 2/5 marks

If A and P are swapped: max 2 /5 marks

✓ answer

(5)



	$4n = \dfrac{\log \frac{A}{P}}{\log(1+i)}$ $4k = \dfrac{\log \frac{30440}{12500}}{\log\left(1+\frac{0,09}{4}\right)}$ $4k = 40,00020365...$ $k = 10$ years	✓ $n = 4k$ ✓ $i = \dfrac{0,09}{4}$ ✓ subs into correct formula ✓ use of logs ✓ answer (5)
7.2.1	30% of R18 480 $= \left(\dfrac{30}{100}\right)(18\ 480)$ = R5 544	✓ answer (1)
7.2.2	$P = \dfrac{x\left[1-(1+i)^{-n}\right]}{i}$ $= \dfrac{5544\left[1-\left(1+\frac{0,08}{12}\right)^{-300}\right]}{\frac{0,08}{12}}$ = R 718 305,71 **OR** $x\left(1+\dfrac{0,08}{12}\right)^{300} = \dfrac{5544\left[\left(1+\frac{0,08}{12}\right)^{300}-1\right]}{\frac{0,08}{12}}$ $x\left(1+\dfrac{0,08}{12}\right)^{300} = 5272490,33...$ x = R 718 305,71 **NOTE:** Incorrect formula: max 2/4 marks If the answer is R 717 550,42 (used n = 299): award 3 /4 marks	✓ $i = \dfrac{0,08}{12}$ ✓ $n = 300$ ✓ substitution ✓ answer (4) ✓ $i = \dfrac{0,08}{12}$ ✓ $n = 300$ ✓ substitution ✓ answer (4) **[12]**



Exemplar November 2014

QUESTION 7

Siphokazi bought a house. She paid a deposit of R102 000, which is equivalent to 12% of the selling price of the house. She obtained a loan from the bank to pay the balance of the selling price. The bank charges her interest of 9% per annum, compounded monthly.

7.1 Determine the selling price of the house. (1)

7.2 The period of the loan is 20 years and she starts repaying the loan one month after it was granted. Calculate her monthly instalment. (4)

7.3 How much interest will she pay over the period of 20 years? Round your answer correct to the nearest rand. (2)

7.4 Calculate the balance of her loan immediately after her 85th instalment. (3)

7.5 She experienced financial difficulties after the 85th instalment and did not pay any instalments for 4 months (that is months 86 to 89). Calculate how much Siphokazi owes on her bond at the end of the 89th month. (2)

7.6 She decides to increase her payments to R8 500 per month from the end of the 90th month. How many months will it take to repay her bond after the new payment of R8 500 per month? (4)
[16]



QUESTION/VRAAG 7

7.1	Selling price / *Verkoopprys* $= \frac{102\ 000}{0{,}12}$ $= 850\ 000$	✓ 850 000 (1)
7.2	$P_v = \frac{x[1-(1+i)^{-n}]}{i}$ $748\ 000 = \frac{x\left[1-\left(1+\frac{0{,}09}{12}\right)^{-240}\right]}{\frac{0{,}09}{12}}$ $x = 6\,729{,}95$ **OR** $F_v = \frac{x[(1+i)^n-1]}{i}$ $748\ 000\left(1+\frac{0{,}09}{12}\right)^{240} = \frac{x\left[\left(1+\frac{0{,}09}{12}\right)^{240}-1\right]}{\frac{0{,}09}{12}}$ $x = 6\,729{,}95$	✓ $Pv = 748\ 000$ ✓ $i = \frac{0{,}09}{12}$ ✓ $n = -240$ ✓ $x = $ R6 729,95 (4) ✓ $748000\,(1+\frac{0{,}09}{12})^{240}$ ✓ $i = \frac{0{,}09}{12}$ ✓ $n = 240$ ✓ $x = $ R6 729,95 (4)
7.3	Total interest paid / *Totale rente betaal* = (6 729,95 x 240) –748 000 = R 867 188	✓ (6 729,95 x 240) ✓ 867 188 (2)
7.4	Balance $= \frac{x[1-(1+i)^{-n}]}{i}$ $= \frac{6729{,}95\left[1-\left(1+\frac{0{,}09}{12}\right)^{-155}\right]}{\frac{0{,}09}{12}}$ $x = 615\ 509{,}74$ **OR**	✓ 6729,95 ✓ $n = -155$ ✓ R615 509,74 (3)



	$A = 748\ 000\left(1+\frac{0,09}{12}\right)^{85}$ $= 1\,411\,663,732$ $F_v = \frac{x[(1+i)^n - 1]}{i}$ $= \frac{6729,95\left[\left(1+\frac{0,09}{12}\right)^{85} - 1\right]}{\frac{0,09}{12}}$ $= 796\,153,962$ Balance of loan $= 1\,411\,663,732 - 796\,153,962$ $= 615\,509,77$ **OR** Balance $= 748\ 000\left(1+\frac{0,09}{12}\right)^{85} - \frac{6729,95\left[\left(1+\frac{0,09}{12}\right)^{85} - 1\right]}{\frac{0,09}{12}}$ $= 615\,509,77$	✓ 1 411 663,732 ✓ $n = 85$ ✓ R615 509,77 (3) ✓ subs of 748 000 and 6729,95 ✓ $n = 85$ ✓ R615 509,77 (3)
7.5	New value of bond: $615\,509,74\left(1+\frac{0,09}{12}\right)^4$ **or** $615\,509,77\left(1+\frac{0,09}{12}\right)^4$ $= 634\,183,81$ $= 634\,183,84$	✓ $R615\,509,74(1+\frac{0,09}{12})^4$ ✓ R634 183,81/ R634 183,84 (2)
7.6	$634\,183,81 = \frac{8\,500\left[1-\left(1+\frac{0,09}{12}\right)^{-n}\right]}{\frac{0,09}{12}}$ $\log(0,44042605) = -n\log\left(1+\frac{0,09}{12}\right)$ $n = 109,74$ $= 110$ months **OR**	✓ $x = 8\,500$ ✓ subs into correct formula ✓ use of logs ✓ answer (4)

	$\mathbf{634\,183{,}81} = \dfrac{8\,500\left[1-\left(1+\dfrac{0{,}09}{12}\right)^{-n}\right]}{\dfrac{0{,}09}{12}}$ $-n = \log_{\left(1+\frac{0{,}09}{12}\right)}(0{,}44042605)$ $n = 109{,}74$ $= 110$ months	✓ $x = 8\,500$ ✓ subs into correct formula ✓ use of logs ✓ answer (4) **[16]**

February 2014

QUESTION 9

Susan buys a car for R350 000. She secures a loan at an interest rate of 7% p.a., compounded monthly. The monthly instalment is R6 300. She pays the first instalment one month after the loan was secured.

9.1 Calculate the effective annual interest rate on the loan. Leave your answer correct to TWO decimal places. (3)

9.2 How many months will it take to repay the loan? (5)

9.3 Calculate the value of the final instalment. (5)

9.4 The value of the car depreciates at i % p.a. After 3 years its value is R252 000. Calculate i. (3)

[16]

QUESTION/*VRAAG* 9

9.1	$1+i_{eff}=\left(1+\frac{i_{nom}}{m}\right)^{m}$ $1+i_{eff}=\left(1+\frac{0,07}{12}\right)^{12}$ $i_{eff}=0,07229008$ $i_{eff}=7,23\%$	✓ substitution into correct formula ✓ $\frac{0,07}{12}$ ✓ answer (3)

9.2	$P_v = \frac{x[1-(1+i)^{-n}]}{i}$ $350\,000 = \frac{6300[1-\left(1+\frac{0.07}{12}\right)^{-n}}{\frac{0,07}{12}}$ $\frac{73}{108} = \left(1+\frac{0.07}{12}\right)^{-n}$ $\log\frac{73}{108} = -n\log\left(1+\frac{0.07}{12}\right)$ $n = 67{,}33938079$ $n = 67{,}34$ months	✓ $i = \frac{0.07}{12}$ ✓ substitution in the correct formula ✓ simplification ✓ use of logs ✓ answer (5)
9.3	$P_v = \frac{x[1-(1+i)^{-n}]}{i}(1+i)$ $P_v = \frac{6\,300[1-(1+\frac{0,07}{12})^{-0,3393....}]}{\frac{0,07}{12}}\left(1+\frac{0,07}{12}\right)$ $P_v = \text{R } 2\,142{,}21$ **OR** Balance outstanding: $= \left[350\,000\left(1+\frac{0,07}{12}\right)^{67} - \frac{6\,300[(1+\frac{0,07}{12})^{67.} - 1]}{\frac{0,07}{12}}\right]\left(1+\frac{0,07}{12}\right)$ $= \text{R } 2\,142{,}21$	✓ $n = -0{,}3393...$ ✓ $i = \frac{0,07}{12}$ ✓ substitution in the correct formula ✓ multiplication by $\left(1+\frac{0,07}{12}\right)$ ✓ answer (5) ✓ $n = -0{,}3393...$ ✓ $i = \frac{0,07}{12}$ ✓ substitution in the correct formula ✓ multiplication by ✓ answer (5)
9.4	$252\,000 = 350\,000(1-i)^3$ $(1-i)^3 = \frac{252\,000}{350\,000}$ $i = 1-\sqrt[3]{\frac{252}{350}}$ $i = 10{,}37\%$	✓ $n = 3$ ✓ substitution in the correct formula ✓ answer (3) **[16]**

November 2014

QUESTION 7

7.1 Exactly five years ago Mpume bought a new car for R145 000. The current book value of this car is R72 500. If the car depreciates by a fixed annual rate according to the reducing-balance method, calculate the rate of depreciation. (3)

7.2 Samuel took out a home loan for R500 000 at an interest rate of 12% per annum, compounded monthly. He plans to repay this loan over 20 years and his first payment is made one month after the loan is granted.

7.2.1 Calculate the value of Samuel's monthly instalment. (4)

7.2.2 Melissa took out a loan for the same amount and at the same interest rate as Samuel. Melissa decided to pay R6 000 at the end of every month. Calculate how many months it took for Melissa to settle the loan. (4)

7.2.3 Who pays more interest, Samuel or Melissa? Justify your answer. (2)

[13]



QUESTION/*VRAAG* 7

7.1	$A = P(1-i)^n$ $72\,500 = 145\,000\,(1-i)^5$ $i = 1-\sqrt[5]{\frac{72500}{145000}}$ $= 0{,}1294\ldots$ $\therefore$ Rate of interest/*Rentekoers* is 12,94 % p.a./*p.j.* **OR/*OF*** $(1-i)^5 = \frac{1}{2}$ $\therefore i = 1-\left(\frac{1}{2}\right)^{\frac{1}{5}}$ $i = 0{,}1294$ $\therefore$ Rate of interest/*Rentekoers* is 12,94 % p.a./*p.j.*	✓substitution/*substitusie* ✓writing in terms of i *herskryf in terme van i* ✓answer/*antwoord* (3) ✓substitution/*substitusie* ✓writing i.t.o i ✓answer (3)
7.2.1	$P = \frac{x\left[1-(1+i)^{-n}\right]}{i}$ $500\,000 = \frac{x\left[1-\left(1+\frac{0{,}12}{12}\right)^{-240}\right]}{\frac{0{,}12}{12}}$ $x = \frac{500000\times\frac{0{,}12}{12}}{\left[1-\left(1+\frac{0{,}12}{12}\right)^{-240}\right]}$ $x = \text{R}5505{,}43$	✓ $i = \frac{0{,}12}{12}$ ✓ $n = 240$ ✓substitution into correct formula ✓answer/*antwoord* (4)

7.2.2	$P = \frac{x\left[1-(1+i)^{-n}\right]}{i}$ $500000 = \frac{6000\left[1-\left(1+\frac{0,12}{12}\right)^{-n}\right]}{\frac{0,12}{12}}$ $\frac{500000}{6000} \times 0,01 = 1-(1,01)^{-n}$ $(1,01)^{-n} = 1-\frac{5}{6}$ $-n = \frac{\log\frac{1}{6}}{\log 1,01}$ $n = 180,07$ $\therefore$ Melissa settles the loan in 181 months	✓6000 ✓substitute into correct formula/*substitusie in korrekte formule* ✓ use of logs/*gebruik van logs* ✓answer/*antwoord* (4)
7.2.3	Samuel He is paying off his loan over a longer period thus more interest will be paid./*Hy betaal sy lening oor 'n langer tydperk af, dus sal hy meer rente betaal.* **OR/*OF*** Samuel He will pay/*Hy betaal* R5505,43 × 240 – R500 000 = R821 303,20 She will pay between/*Sy sal tussen* R580 000 and/*en* R586 000,00 *betaal.*	✓Samuel ✓reason/*rede* (2) ✓Samuel ✓reason/*rede* (2) **[13]**



February 2015

QUESTION 7

7.1 Nomsa started working on 1 January 1970. At the end of January 1970 and at the end of each month thereafter, she deposited R400 into an annuity fund. She continued doing this until she retired on 31 December 2013.

7.1.1 Determine the total amount of money that she paid into the fund. (2)

7.1.2 The interest rate on this fund was 8% p.a., compounded monthly. Calculate the value of the fund at the time that she retired. (5)

7.1.3 On 1 January 2014 Nomsa invested R2 million in an account paying interest at 10% p.a. compounded monthly. Nomsa withdraws a fixed amount from this account at the end of each month, starting on 31 January 2014. If Nomsa wishes to make monthly withdrawals from this account for 25 years, calculate the maximum amount she could withdraw at the end of each month. (4)

7.2 For each of the three years from 2010 to 2012 the population of town X decreased by 8% per year and the population of town Y increased by 12% per year.

At the end of 2012 the populations of these two towns were equal.

Determine the ratio of the population of town X (call it P_X) to the population of town Y (call it P_Y) at the beginning of 2010. (4)

[15]



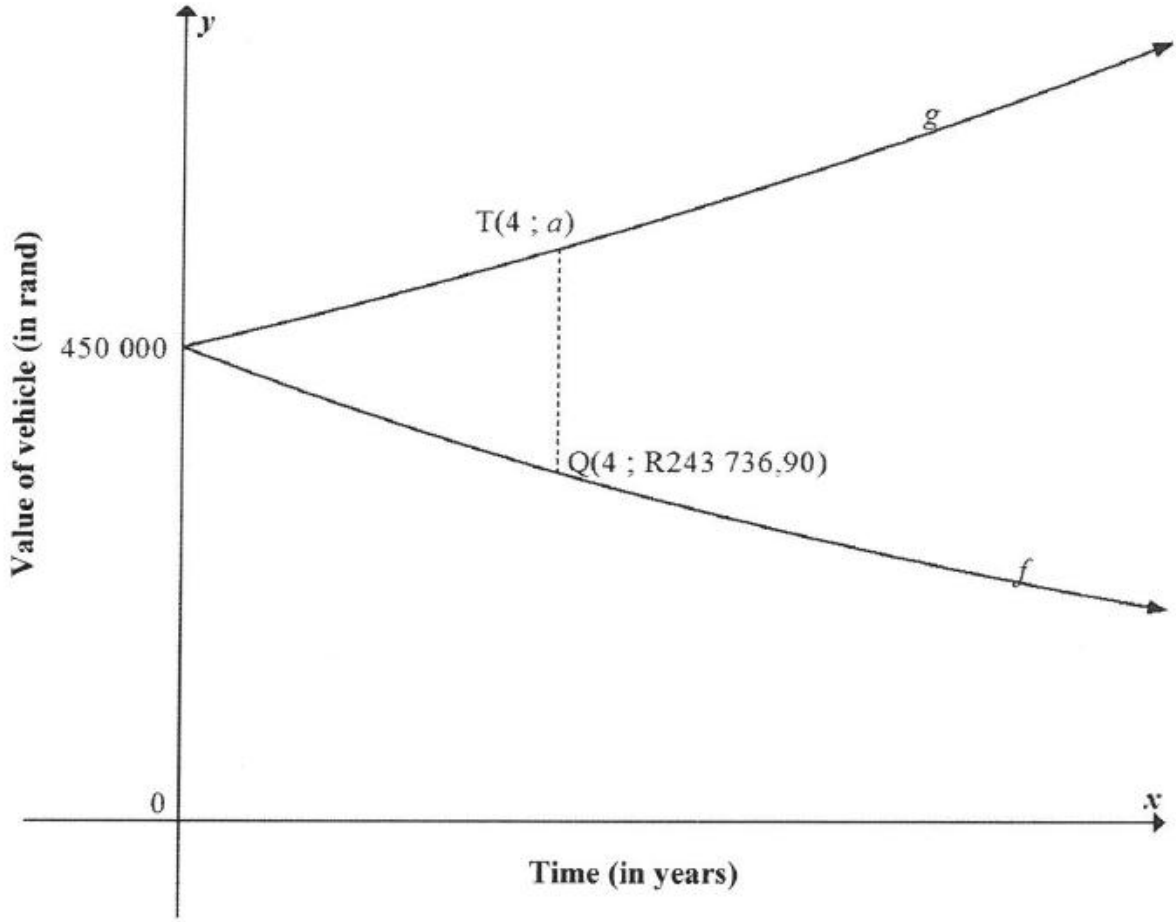
QUESTION/*VRAAG* 7

7.1.1	$R400\times(44\times12)$ $=R211200$	✓ $R400\times(44\times12)$ ✓ $R211200$ (2)
7.1.2	$F=\frac{x\left[(1+i)^n-1\right]}{i}$ $=\frac{400\left[\left(1+\frac{0,08}{12}\right)^{528}-1\right]}{\frac{0,08}{12}}$ $=R1\ 943\ 524,42$	✓ $x=400$ ✓ $n=528$ ✓ $i=\frac{0,08}{12}$ ✓ substitution into correct formula/*substitusie in korrekte formule* ✓ answer/*antwoord* (5)
7.1.3	$P=\frac{x[1-(1+i)^{-n}]}{i}$ $2000000=\frac{x\left[1-\left(1+\frac{0,1}{12}\right)^{-300}\right]}{\frac{0,1}{12}}$ $x=R18\ 174,01$ **OR/*OF*** $2000000\left(1+\frac{0,1}{12}\right)^{300}=\frac{x\left(\left(1+\frac{0,1}{12}\right)^{300}-1\right)}{\frac{0,1}{12}}$ $x=R18174,01$	✓ P = 2000000 ✓ $n=300$ and/*en* $i=\frac{0,1}{12}$ ✓ substituting into correct formula/*substitusie in korrekte formule* ✓ answer/*antwoord* (4) ✓ P = 2000000 ✓ $n=300$ and/*en* $i=\frac{0,1}{12}$ ✓ equating/stel gelyk ✓ answer/*antwoord* (4)
7.2	Let P_X and P_Y be the populations of the two towns at the beginning of 2010./*Laat* P_X *en* P_Y *die bevolkings wees van die twee dorpe aan die begin van 2010.* $A_X=A_Y$ $P_X(1-0,08)^3=P_Y(1+0,12)^3$ $\frac{P_X}{P_Y}=\frac{(1+0,12)^3}{(1-0,08)^3}$ $=\frac{1,404...}{0,778...}$ $=1,8:1$	✓ equating/*stel gelyk* ✓ $A_X=P_X(1-0,08)^3$ ✓ $A_Y=P_Y(1+0,12)^3$ ✓ answer/*antwoord* (4) **[15]**

November 2015

QUESTION 7

The graph of f shows the book value of a vehicle x years after the time Joe bought it.
The graph of g shows the cost price of a similar new vehicle x years later.



7.1 How much did Joe pay for the vehicle? (1)

7.2 Use the reducing-balance method to calculate the percentage annual rate of depreciation of the vehicle that Joe bought. (4)

7.3 If the average rate of the price increase of the vehicle is 8,1% p.a., calculate the value of a. (3)

7.4 A vehicle that costs R450 000 now, is to be replaced at the end of 4 years. The old vehicle will be used as a trade-in. A sinking fund is created to cover the replacement cost of this vehicle. Payments will be made at the end of each month. The first payment will be made at the end of the 13th month and the last payment will be made at the end of the 48th month. The sinking fund earns interest at a rate of 6,2% p.a., compounded monthly.

Calculate the monthly payment to the fund. (5)
[13]

QUESTION/*VRAAG* 7

7.1	R450 000	✓answer (1)
7.2	$A = P(1-i)^n$ $f(x) = 450000(1-i)^x$ $243\ 736{,}90 = 450000(1-i)^4$ $i = 1 - \sqrt[4]{\frac{243\ 736{,}90}{450000}}$ $i = 0{,}1421$ The rate of depreciation is 14,21% p.a. *Die waardeverminderingskoers is 14,21% p.j.*	✓substitution of 450 000 into correct formula ✓substitution of (4; 243 736,90) into correct formula ✓making i the subject ✓answer (4)
7.3	At T : $A = P(1+i)^n$ $g(x) = 450000(1+i)^x$ $a = 450000(1+0{,}081)^4$ $= R614490{,}66$	✓$i = 0{,}081$ & $n = 4$ ✓correct substitution into formula ✓answer (3)
7.4	Future Value $= R614\,490{,}66 - R243\,736{,}90$ $= R370\,753{,}76$ Let x be the value of monthly payment $F_v = \frac{x\left[(1+i)^n - 1\right]}{i}$ $370753{,}76 = \frac{x\left[\left(1+\frac{0.062}{12}\right)^{36} - 1\right]}{\frac{0.062}{12}}$ $x = R9397{,}11$	✓R370 753,76 ✓$i = \frac{0{,}062}{12}$ ✓$n = 36$ ✓substitution into correct formula ✓answer (5) **[13]**

QUESTION 7

7.1 Diane invests a lump sum of R5 000 in a savings account for exactly 2 years. The investment earns interest at 10% p.a., compounded quarterly.

7.1.1 What is the quarterly interest rate for Diane's investment? (1)

7.1.2 Calculate the amount in Diane's savings account at the end of the 2 years. (3)

7.2 Motloi inherits R800 000. He invests all of his inheritance in a fund which earns interest at a rate of 14% p.a., compounded monthly. At the end of each month he withdraws R10 000 from the fund. His first withdrawal is exactly one month after his initial investment.

7.2.1 How many withdrawals of R10 000 will Motloi be able to make from this fund? (5)

7.2.2 Exactly four years after his initial investment Motloi decides to withdraw all the remaining money in his account and to use it as a deposit towards a house.

(a) What is the value of Motloi's deposit, to the nearest rand? (4)

(b) Motloi's deposit is exactly 30% of the purchase price of the house. What is the purchase price of the house, to the nearest rand? (1)

[14]



QUESTION/*VRAAG* 7

7.1.1	Quarterly interest rate/*Kwartaallikse rentekoers* $= \frac{10\%}{4}$ $= 2{,}5\%$	✓answer (1)
7.1.2	$A = P(1+i)^n$ $= 5000\left(1+\frac{2{,}5}{100}\right)^{2\times 4}$ $= \text{R}6\,092{,}01$	✓$n = 8$ ✓$5000\left(1+\frac{2{,}5}{100}\right)^{2\times 4}$ ✓answer (3)
7.2.1	$P_v = \frac{x\left[1-(1+i)^{-n}\right]}{i}$ $800\,000 = \frac{10\,000\left[1-\left(1+\frac{0{,}14}{12}\right)^{-n}\right]}{\frac{0{,}14}{12}}$ $\frac{800\,000}{10\,000}\times\frac{0{,}14}{12} = 1-\left(1+\frac{0{,}14}{12}\right)^{-n}$ $\left(1+\frac{0{,}14}{12}\right)^{-n} = 1-\frac{800\,000}{10\,000}\times\frac{0{,}14}{12}$ $-n = \frac{\log\left[1-\frac{800000\times\frac{0{,}14}{12}}{10000}\right]}{\log\left(1+\frac{0{,}14}{12}\right)}$ $n = 233{,}4699962$ Motloi can make 233 withdrawals of R10 000./*Motloi kan 233 onttrekkings van R10 000 maak.*	✓$i = \frac{0{,}14}{12}$ ✓substitute into present value formula ✓$\left(1+\frac{0{,}14}{12}\right)^{-n} = 1-\frac{800\,000}{10\,000}\times\frac{0{,}14}{12}$ ✓ use of logs ✓233 (5)
7.2.2 (a)	$\text{A - F}_\text{v} = 800\,000\left(1+\frac{0{,}14}{12}\right)^{48} - \frac{10\,000\left[\left(1+\frac{0{,}14}{12}\right)^{48}-1\right]}{\frac{0{,}14}{12}}$ $= 1\ 396\ 005{,}54 \quad - \ 638\ 577{,}36$ $= \text{R}757\,428$ **OR/*OF***	✓$n = 48$ in both formulae ✓$i = \frac{0{,}14}{12}$ in both formulae ✓ substitution into both formulae ✓answer (4)

	$P_v = \frac{x\left[1-(1+i)^{-n}\right]}{i}$ $= \frac{10000\left[1-\left(1+\frac{0,14}{12}\right)^{-185,4699962...}\right]}{\frac{0,14}{12}}$ $= \text{R}757\,428$	✓$n = -185,46996...$ ✓$i = \frac{0,14}{12}$ ✓ $\frac{10000\left[1-\left(1+\frac{0,14}{12}\right)^{-185,4699962...}\right]}{\frac{0,14}{12}}$ ✓answer (4)
7.2.2 (b)	Let the purchase price of the house be y./*Laat die koopprys van die huis y wees.* $\frac{757\,428}{y} = 30\%$ $757\,428 = 0,3y$ $y = \frac{757\,428}{0,3}$ $= \text{R}2\,524\,760$ **OR/*OF*** Let the purchase price of the house be y./*Laat die koopprys van die huis y wees.* $y = \frac{757\,428}{30} \times 100$ $= \text{R}2\,524\,760$	✓answer (1) ✓answer (1) **[14]**



November 2016

QUESTION 7

On 1 June 2016 a bank granted Thabiso a loan of R250 000 at an interest rate of 15% p.a. compounded monthly, to buy a car. Thabiso agreed to repay the loan in monthly instalments commencing on 1 July 2016 and ending 4 years later on 1 June 2020. However, Thabiso was unable to make the first two instalments and only commenced with the monthly instalments on 1 September 2016.

7.1 Calculate the amount Thabiso owed the bank on 1 August 2016, a month before he paid his first monthly instalment. (2)

7.2 Having paid the first monthly instalment on 1 September 2016, Thabiso will still pay his last monthly instalment on 1 June 2020. Calculate his monthly instalment. (4)

7.3 If Thabiso paid R9 000 as his monthly instalment starting on 1 September 2016, how many months sooner will he repay the loan? (5)

7.4 If Thabiso paid R9 000 as a monthly instalment starting on 1 September 2016, calculate the final instalment to repay the loan. (4)

[15]



QUESTION/*VRAAG* 7

7.1	$A = P(1+i)^n$ $= 250000\left(1+\frac{0,15}{12}\right)^2$ $= R256\ 289,06$	✓substituting i and n values in correct formula ✓ answer (2)
7.2	$P = \frac{x\left[1-(1+i)^{-n}\right]}{i}$ $256\ 289,06 = \frac{x\left[1-\left(1+\frac{0,15}{12}\right)^{-46}\right]}{\frac{0,15}{12}}$ $3203,6133 = x\left[1-\left(1+\frac{0,15}{12}\right)^{-46}\right]$ $x = R\ 7\ 359,79$ per month **OR/*OF*** $250\,000 = \frac{x\left(1+\frac{0,15}{12}\right)^{-2}\left[1-\left(1+\frac{0,15}{12}\right)^{-46}\right]}{\frac{0,15}{12}}$ $x = R\ 7\ 359,79$	✓ $i = \frac{0,15}{12}$ ✓ $n = 46$ ✓ substitution into correct formula ✓ answer (4) ✓ $i = \frac{0,15}{12}$ ✓ $n = 46$ ✓ substitution into correct formula ✓ answer (4)
7.3	$256\ 289,06 = \frac{9\ 000\left[1-\left(1+\frac{0,15}{12}\right)^{-n}\right]}{\frac{0,15}{12}}$ $\left(1+\frac{0,15}{12}\right)^{-n} = 0,6440429722$ $-n\log\left(1+\frac{0,15}{12}\right) = \log 0,6440429722$ $n = 35,41872568$ months/ *maande* $\therefore$ 36 payments are required $\therefore$ 36 *paaiemente moet betaal word* $\therefore$ Thabiso will pay his loan off 10 months sooner./Thabiso *los sy lening 10 maande vroeër af.* **OR/*OF***	✓$x = 9\ 000$ ✓ substitute into correct formula ✓ use of logs ✓ n =35,42 ✓ 10 months (5)

	$256289{,}06\left(1+\frac{0{,}15}{12}\right)^n = \frac{9000\left[\left(1+\frac{0{,}15}{12}\right)^n - 1\right]}{\frac{0{,}15}{12}}$ $3203{,}61325\left(1+\frac{0{,}15}{12}\right)^n = 9000\left(1+\frac{0{,}15}{12}\right)^n - 9000$ $9000 = 5796{,}38675\left(1+\frac{0{,}15}{12}\right)^n$ $n = \log_{\left(1+\frac{0{,}15}{12}\right)} 1{,}5523691425$ $n = 35{,}41872568$ $\therefore$ 36 payments are required $\therefore$ 36 *paaiemente moet betaal word* $\therefore$ Thabiso will pay his loan off 10 months sooner./Thabiso *los sy lening 10 maande vroeër af.*	✓ 9 000 ✓ substitute into correct formula ✓ use of logs ✓ n =35,42 ✓ 10 months (5)
7.4	The balance of his loan after the 35th payment was made: *Die balans van sy lening nadat die 35ste paaiement betaal is:* $\text{Balance} = 256289{,}06\left(1+\frac{0{,}15}{12}\right)^{35} - \frac{9000\left(\left(1+\frac{0{,}15}{12}\right)^{35} - 1\right)}{\frac{0{,}15}{12}}$ $= \text{R } 3\,735{,}45$ $\text{Final instalment} = 3\,735{,}45\left(1+\frac{0{,}15}{12}\right)$ $= \text{R } 3\,782{,}14$	✓ $256289{,}06\left(1+\frac{0{,}15}{12}\right)^{35}$ ✓ $\frac{9000\left(\left(1+\frac{0{,}15}{12}\right)^{35} - 1\right)}{\frac{0{,}15}{12}}$ ✓ $3\,735{,}45\left(1+\frac{0{,}15}{12}\right)$ ✓ answer (4)
	OR/OF $P = \frac{x\left[1-(1+i)^{-n}\right]}{i}$ Final instalment $= \frac{9\,000\left[1-\left(1+\frac{0{,}15}{12}\right)^{-0{,}41872568}\right]}{\frac{0{,}15}{12}}\left(1+\frac{0{,}15}{12}\right)$ $= \text{R } 3\,782{,}14$	✓ 0,41872568 ✓ $\frac{9\,000\left[1-\left(1+\frac{0{,}15}{12}\right)^{-0{,}41872568}\right]}{\frac{0{,}15}{12}}$ ✓ $\times\left(1+\frac{0{,}15}{12}\right)$ ✓ answer (4)
	OR/OF	

	$\text{Balance} = 256289{,}06\left(1+\frac{0{,}15}{12}\right)^{36} - \frac{9000\left(\left(1+\frac{0{,}15}{12}\right)^{36}-1\right)}{\frac{0{,}15}{12}}$ $= \text{R } -5\,217{,}86$ $\text{Final payment} = 9\,000 - 5217{,}86$ $= \text{R } 3\,782{,}14$	✓ $256289{,}06\left(1+\frac{0{,}15}{12}\right)^{36}$ ✓ $\frac{9000\left(\left(1+\frac{0{,}15}{12}\right)^{36}-1\right)}{\frac{0{,}15}{12}}$ ✓ $9\,000 - 5217{,}86$ ✓ answer (4) **[15]**

QUESTION 6

6.1 On the 2nd day of January 2015 a company bought a new printer for R150 000.

- The value of the printer decreases by 20% annually on the reducing-balance method.
- When the book value of the printer is R49 152, the company will replace the printer.

6.1.1 Calculate the book value of the printer on the 2nd day of January 2017. (3)

6.1.2 At the beginning of which year will the company have to replace the printer? Show ALL calculations. (4)

6.1.3 The cost of a similar printer will be R280 000 at the beginning of 2020. The company will use the R49 152 that it will receive from the sale of the old printer to cover some of the costs of replacing the printer. The company set up a sinking fund to cover the balance. The fund pays interest at 8,5% per annum, compounded quarterly. The first deposit was made on 2 April 2015 and every three months thereafter until 2 January 2020. Calculate the amount that should be deposited every three months to have enough money to replace the printer on 2 January 2020. (4)

6.2 Lerato wishes to apply for a home loan. The bank charges interest at 11% per annum, compounded monthly. She can afford a monthly instalment of R9 000 and wants to repay the loan over a period of 15 years. She will make the first monthly repayment one month after the loan is granted. Calculate, to the nearest thousand rand, the maximum amount that Lerato can borrow from the bank. (5)

[16]



QUESTION/*VRAAG* 6

6.1.1	$A = 150\,000(1-0,2)^2$ $= R96\,000$	✓ $n = 2$ ✓ 150 000 in correct formula ✓ 96 000 (3)
6.1.2	$150\,000(1-0,2)^n = 49\,152$ $(0,8)^n = \frac{1024}{3125}$ $n\log(0,8) = \log\frac{1024}{3125}$ $n = 5$ The machine will need to be replaced at the beginning of 2020 / *Masjien moet aan die begin van 2020 vervang word* **OR / OF** $150\,000(1-0,2)^n = 49\,152$ $(0,8)^n = \frac{1024}{3125}$ $n = \log_{0,8}\frac{1024}{3125}$ $n = 5$ The machine will need to be replaced at the beginning of 2020 / *Masjien moet aan die begin van 2020 vervang word*	✓ $150\,000(1-0,2)^n = 49\,152$ ✓ $n\log(0,8) = \log\frac{1024}{3125}$ ✓ $n = 5$ ✓ 2020 (4) ✓ $150\,000(1-0,2)^n = 49\,152$ ✓ $n = \log_{0,8}\frac{1024}{3125}$ ✓ $n = 5$ ✓ 2020 (4)



6.1.3	R280 000 – R49 152 =R230 848 $230\ 848 = \dfrac{x\left[\left(1+\dfrac{0,085}{4}\right)^{20}-1\right]}{\dfrac{0,085}{4}}$ $x = \text{R}9\,383,26$	✓ R230 848 ✓ $i = \dfrac{0,085}{4} = 0,02125$ and $n = 20$ ✓ subs into correct formula ✓ R 9 383,26 (4)
6.2	$P_v = \dfrac{x[1-(1+i)^{-n}]}{i}$ $= \dfrac{9\,000\left[1-\left(1+\dfrac{0,11}{12}\right)^{-180}\right]}{\dfrac{0,11}{12}}$ = R791 837,43 Lerato qualifies for a loan of R 791 000 under the given conditions / *Lerato kwalifiseer vir 'n lening van* R 791 000 *gegewe die kondisies*	✓ $i = \dfrac{0,11}{12}$ ✓ $n = 180$ ✓ substitution correct formula ✓ R791 837,43 ✓ R791 000 (5) **[16]**